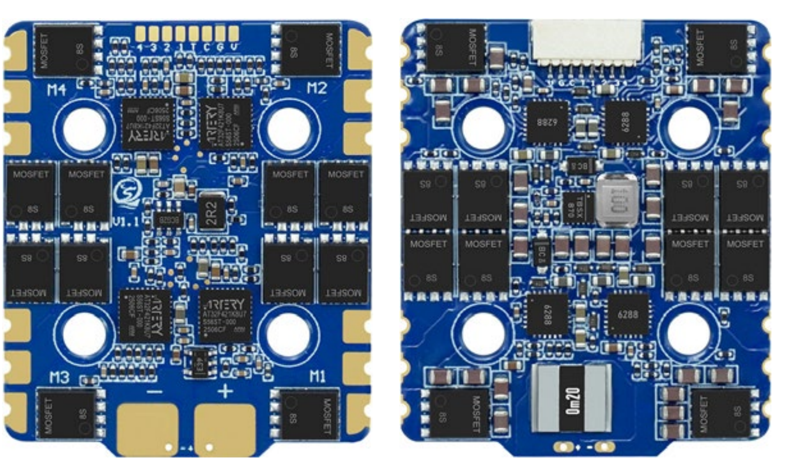
V1.0

SEQURE

SEQURE Blueson A1
电调说明书

感谢您购买本产品！请您在使用前仔细阅读说明书，严格遵守规定的操作说明，使用者需具备一定电子基础和良好的焊接技术，错误的使用可能造成人身伤害和设备损坏。我们不承担因使用本产品或擅自对产品进行改造而引起的任何责任，包括但不限于对附带损失或间接损失的赔偿责任。

Thanks for buying this product! Please read the instructions carefully before use and strictly abide by the specified operating instructions. Users must have a certain electronic foundation and good welding skills. Wrong use may cause personal injury and equipment damage. We do not assume any liability arising from the use of this product or unauthorized modifications to the product, including but not limited to liability for incidental or indirect losses.



1. English instruction manual .... 01 - 03
2. 中文说明书 .... 04 - 06

CE FC

1 PRODUCT DESCRIPTION

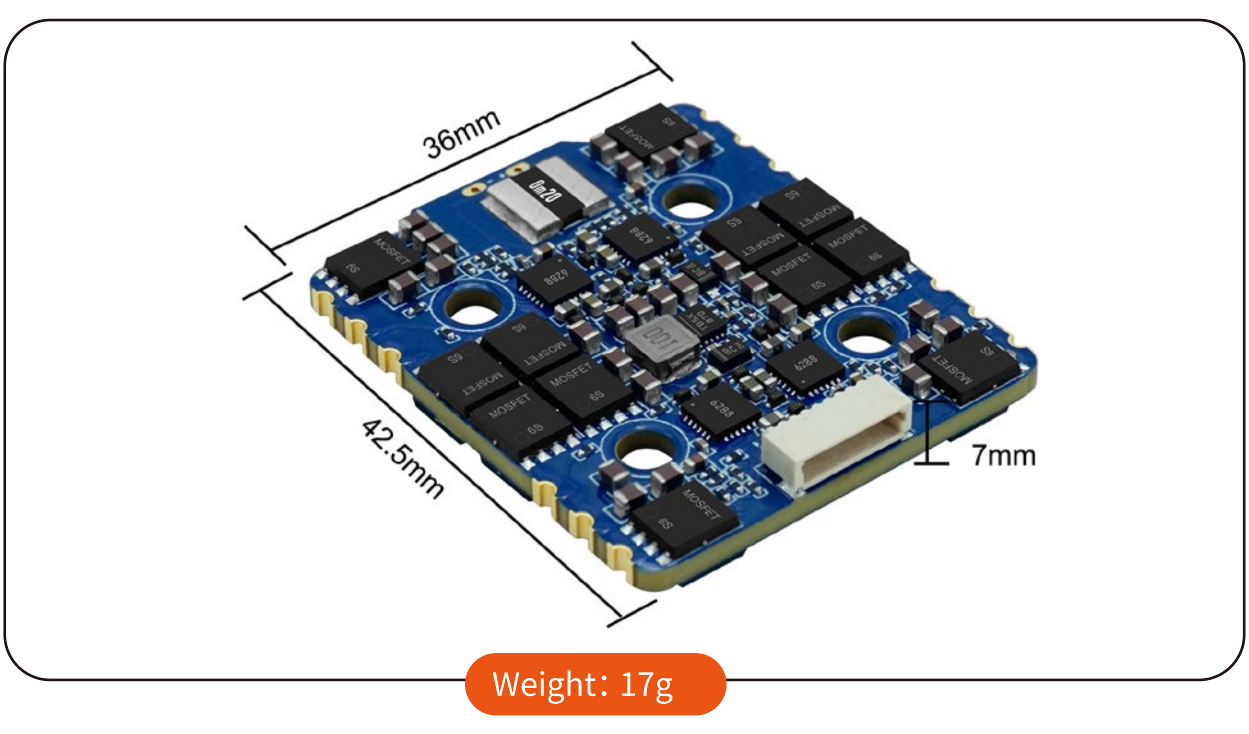
- AT32F421 high-performance processor, using BLHeli_32 | AM32 firmware, 2-8S wide voltage input, single-channel continuous current 70A, peak current up to 150A.
- Built-in galvanometer, supports telemetry return, and updates motor speed, temperature, current and other data in real time.
- High-quality high-power MOSFET, ultra-high efficiency conversion, reducing losses and extending flight time.
- High copper thickness multi-layer PCB design increases over-current capability, efficient heat dissipation performance, and is not afraid of load pressure.
- Precise switch interval time control, speeding up the response speed of the ESC, smooth control, and delicate feel.
- PWM frequency supports 8KHz-128KHz, and excellent circuit design solves the problem of out-of-synchronization.
- Innovative design, motor wiring layout is more beautiful and safe, while reducing welding difficulty.
- 20*20mm installation hole spacing design, compact structure, adaptable to various types of racks, no pressure during installation.

2 PRODUCT PARAMETERS

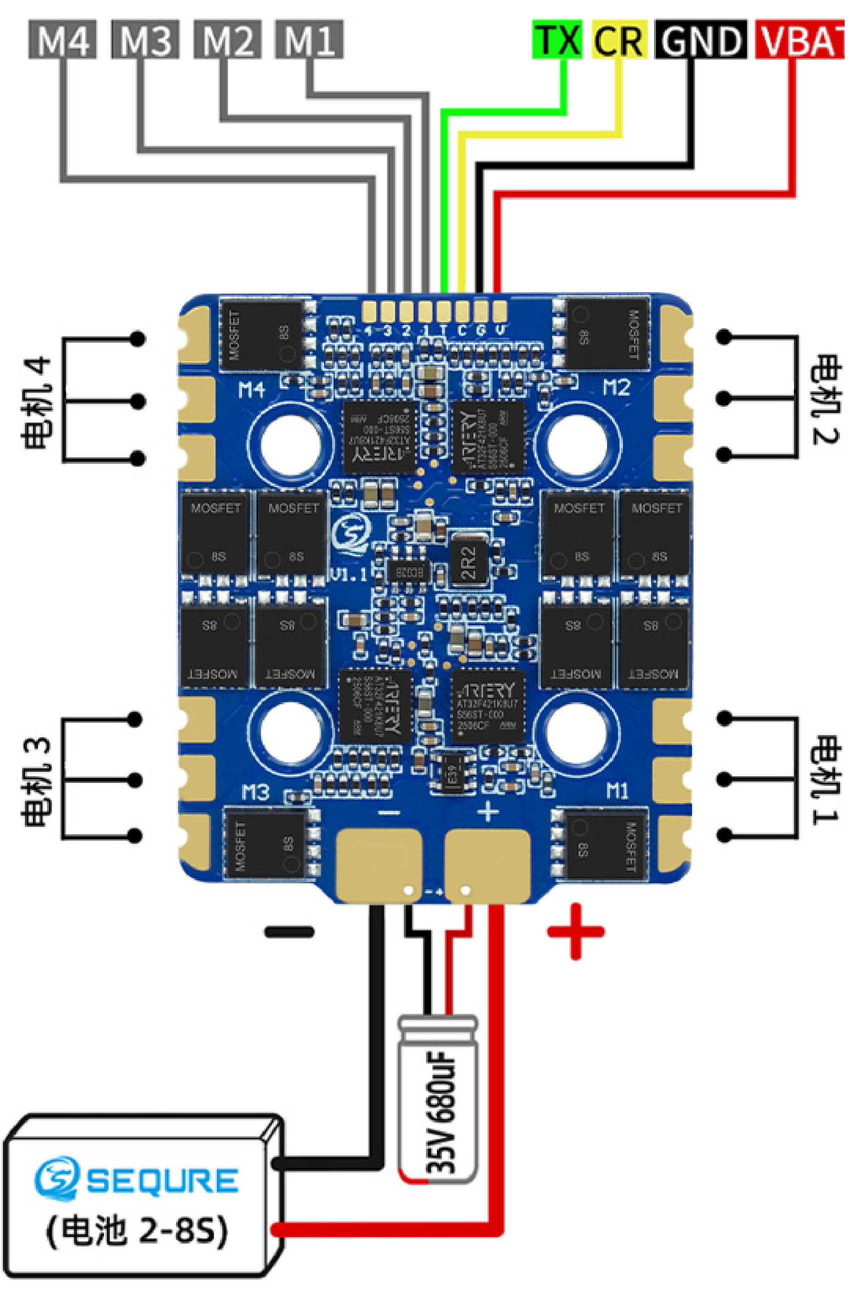
Model	Blueson A1
Processor	AT32F421
PWM Frequency	8KHz-128KHz
Working Voltage	2-6S / 2-8S Lipo/LiHV
Continuous Current	6S 65A / 8S 70A
Peak Current	6S 120A / 8S 150A
Support Protocol	Dshot600/300/150, Oneshot,Multishot,PWM

Firmware Version	AM32 or BLHeli_32
Telemetry Return	YES
Galvanometer	YES
BEC	NO
Mounting Hole Spacing	20*20mm, Φ4mm/M3
Product Size	42.5*36*7mm (L*W*h)
Product Weight	19g

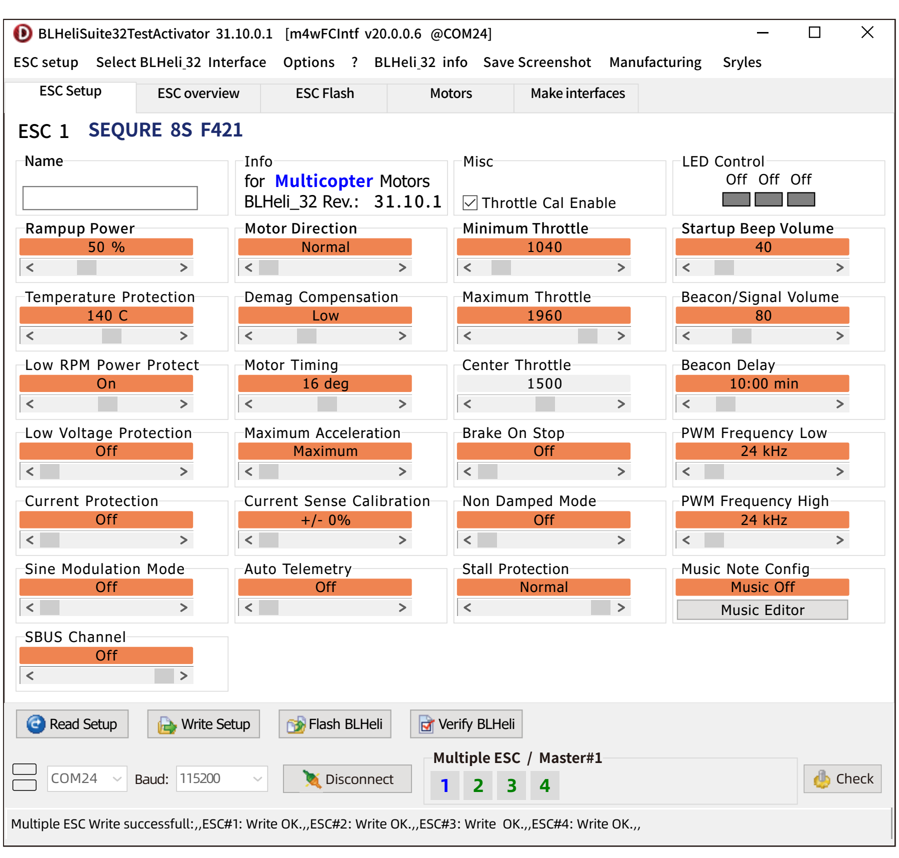
3 SIZE AND WEIGHT



4 WIRING DIAGRAM



5 ESC SETTING

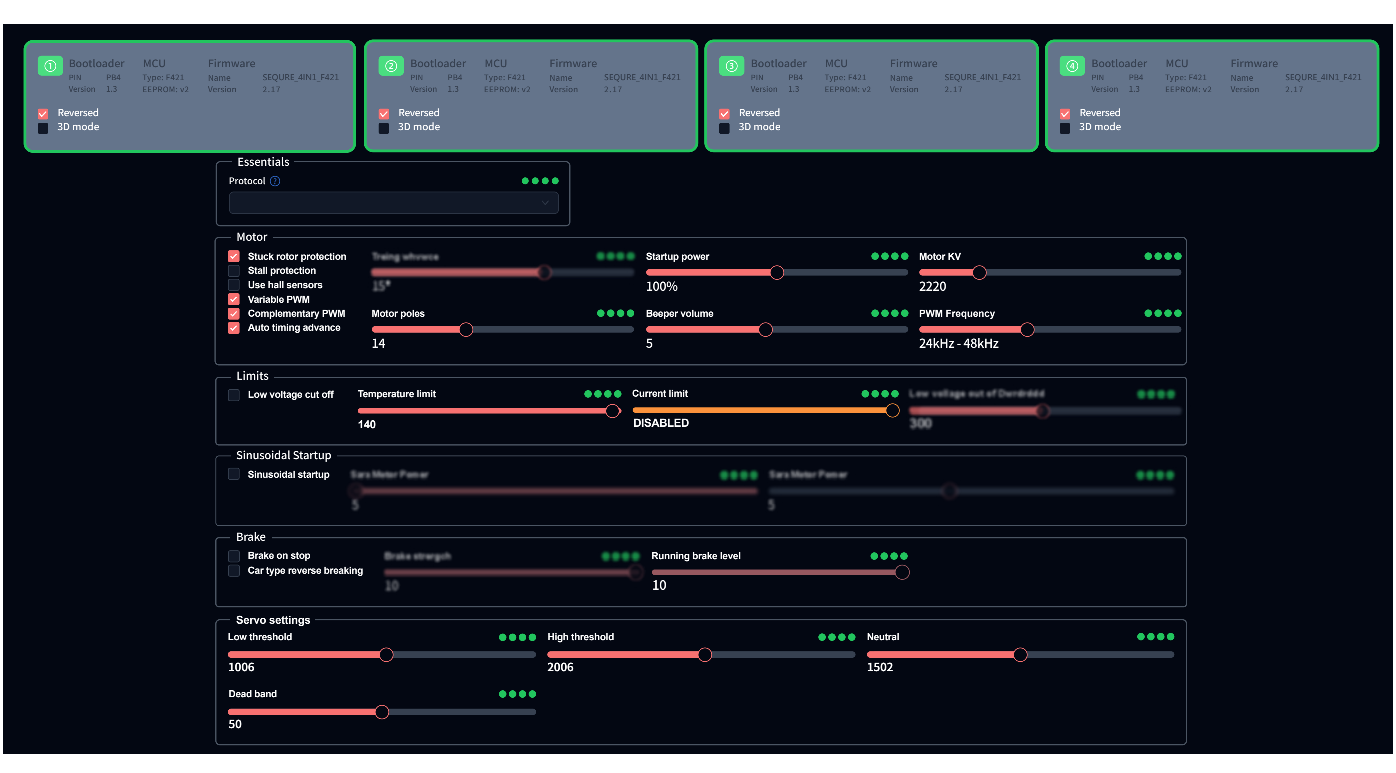


ESC 1 SEQURE 8S F421

Name:
Info: for Multicopter Motors, BLHeli_32 Rev.: 31.10.1
Misc: ☑ Throttle Cal Enable
LED Control: Off Off Off
Rampup Power: 50 %
Motor Direction: Normal
Minimum Throttle: 1040
Startup Beep Volume: 40
Temperature Protection: 140 C
Demag Compensation: Low
Maximum Throttle: 1960
Beacon/Signal Volume: 80
Low RPM Power Protect: On
Motor Timing: 16 deg
Center Throttle: 1500
Beacon Delay: 10:00 min
Low Voltage Protection: Off
Maximum Acceleration: Maximum
Brake On Stop: Off
PWM Frequency Low: 24 kHz
Current Protection: Off
Current Sense Calibration: +/- 0%
Non Damped Mode: Off
PWM Frequency High: 24 kHz
Sine Modulation Mode: Off
Auto Telemetry: Off
Stall Protection: Normal
Music Note Config: Music Off
SBUS Channel: Off
COM24 Baud: 115200
Multiple ESC / Master#1: 1 2 3 4

Multiple ESC Write successfull:,,ESC#1: Write OK.,,ESC#2: Write OK.,,ESC#3: Write OK.,,ESC#4: Write OK.,,

BLHeli_32 firmware visual programming
parameter adjustment interface



ESC ①–④: Bootloader PIN PB4, Version 1.3; MCU Type: F421, EEPROM: v2; Firmware Name SEQURE_4IN1_F421, Version 2.17; ☑ Reversed; ☐ 3D mode

Protocol:

☑ Stuck rotor protection
☐ Stall protection
☐ Use hall sensors
☑ Variable PWM
☑ Complementary PWM
☑ Auto timing advance

Startup power: 100%
Motor KV: 2220
Motor poles: 14
Beeper volume: 5
PWM Frequency: 24kHz - 48kHz

☐ Low voltage cut off
Temperature limit: 140
Current limit: DISABLED

☐ Sinusoidal startup

☐ Brake on stop
☐ Car type reverse breaking
Running brake level: 10

Low threshold: 1006
High threshold: 2006
Neutral: 1502
Dead band: 50

AM32 firmware visual programming
parameter adjustment interface
Website：am32.ca

6 WARRANTY TERMS

1. For those performance faults caused by non-human factors within 3 months from the date of purchase, we provide repair & maintenance service for free. After 3 months, no labor fee will be charged for repairs. If you would like to replace parts, you must pay for the cost.
2. All man-made damage, faults caused by self disassembly & improper use, and all consumables like soldering tips are not covered by the warranty.

7 SAFETY PRECAUTIONS ⚠

- Please do not flash unofficial firmware to avoid damage to the ESC.
- Please do not use it beyond the ESC operating voltage and current range.
- Please check and make sure the positive and negative polarity are correct before powering on.
- Please use the product in compliance with local regulations and ensuring safety.
- Please use accessories that meet safety standards and store them out of reach of children.

www.sequremall.com
info@sequremall.com

1 产品描述

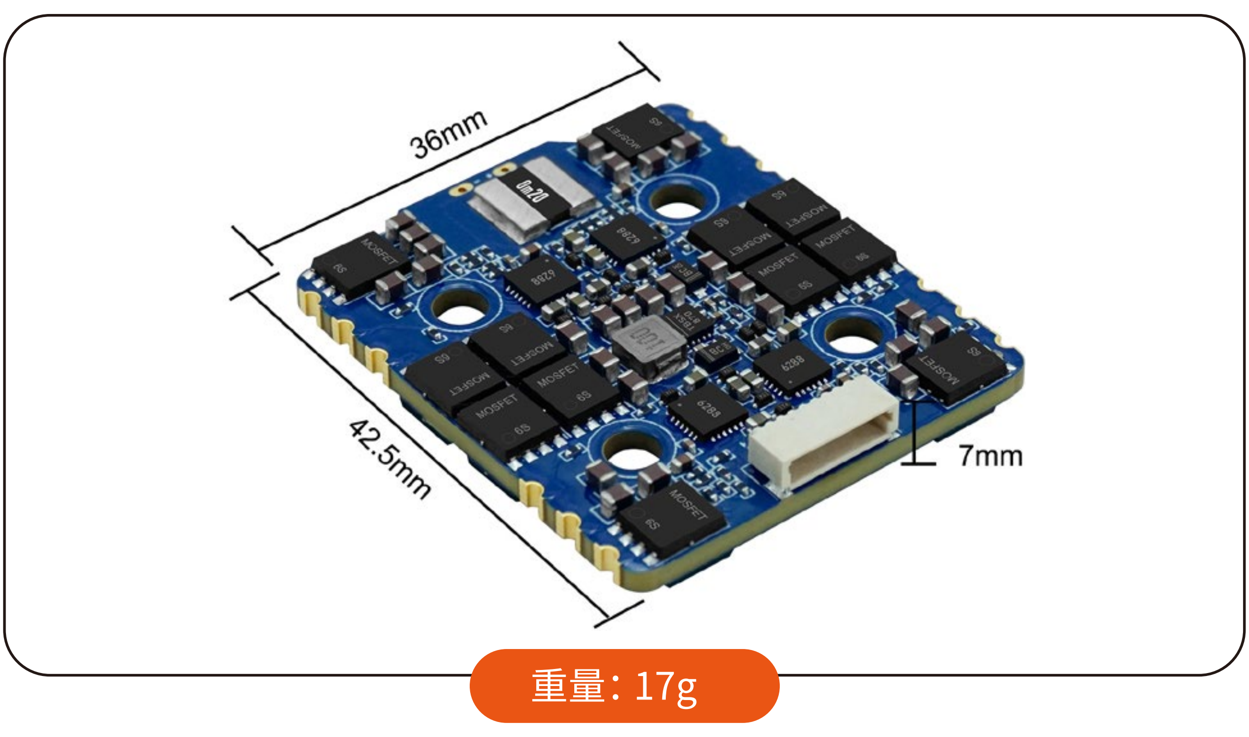
- ●AT32F421高性能处理器，使用BLHeli_32 | AM32 固件，2-8S宽电压输入，单路持续电流70A，峰值电流高达150A。
- ●内置电流计，支持遥测回传，实时更新电机转速，温度，电流等数据。
- ●优质的高功率MOSFET，超高的效率转换，降低损耗，延长飞行时间。
- ●高铜厚多层PCB设计，增大过流能力，高效的散热性能，无惧负载压力。
- ●精准开关间隔时间控制，加快电调的响应速度，控制流畅，手感细腻。
- ●PWM频率支持8KHz-128KHz，优异的电路设计，解决失步问题。
- ●创新设计，电机走线布局更加美观安全，同时降低焊接难度。
- ●20*20mm 安装孔距设计，结构紧凑，适配多种类型机架，装机无压力。

2 产品参数

型号名称	Blueson A1
处理器	AT32F421
PWM频率	8KHz-128KHz
工作电压	2-6S / 2-8S Lipo/LiHV
持续电流	6S 65A / 8S 70A
峰值电流	6S 120A / 8S 150A
支持协议	Dshot600/300/150,Oneshot, Multishot,PWM

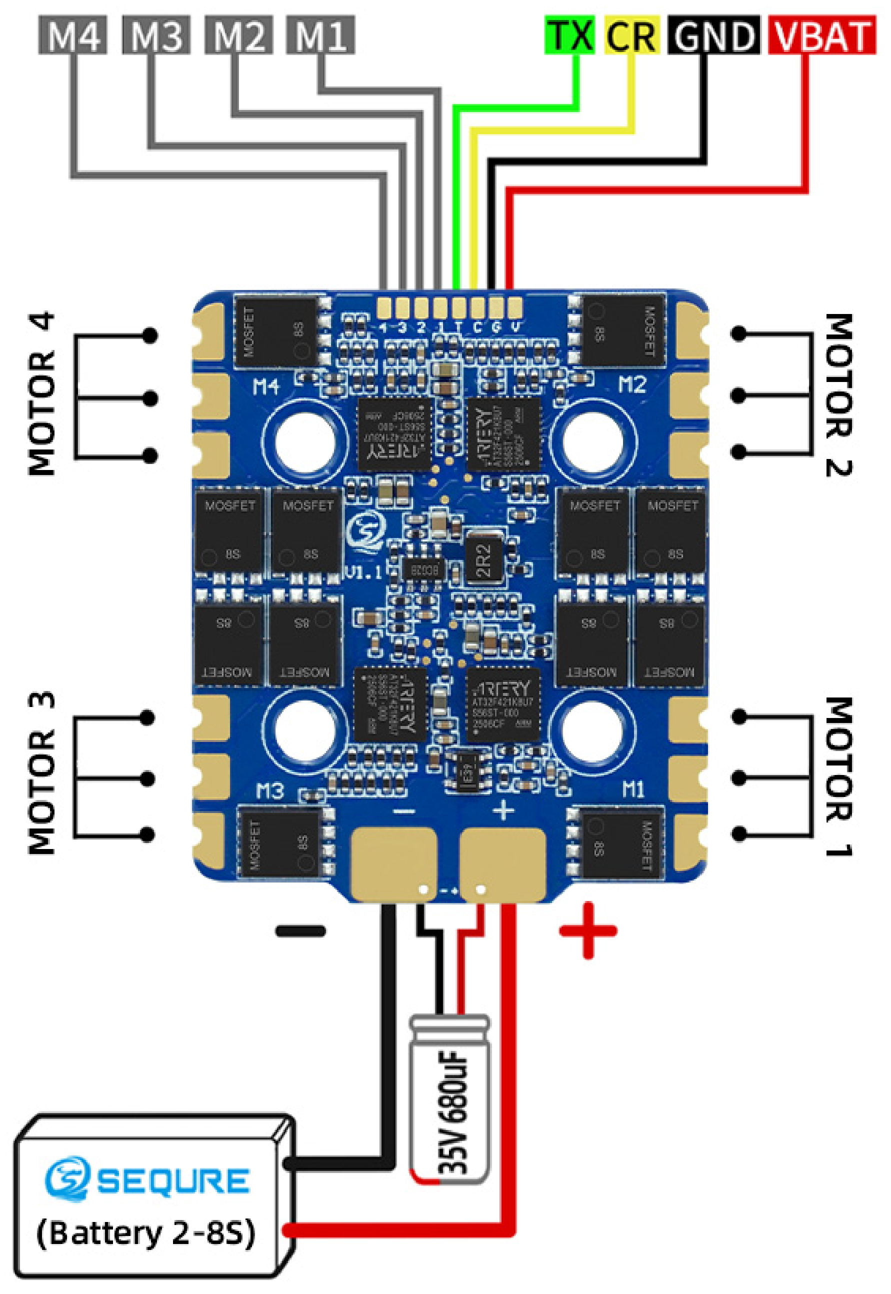
固件版本	AM32 or BLHeli_32
遥测回传	YES
电流计	YES
BEC	NO
安装孔距	20*20mm, Φ4mm/M3
产品尺寸	42.5*36*7mm (L*W*h)
产品重量	19g

3 尺寸重量

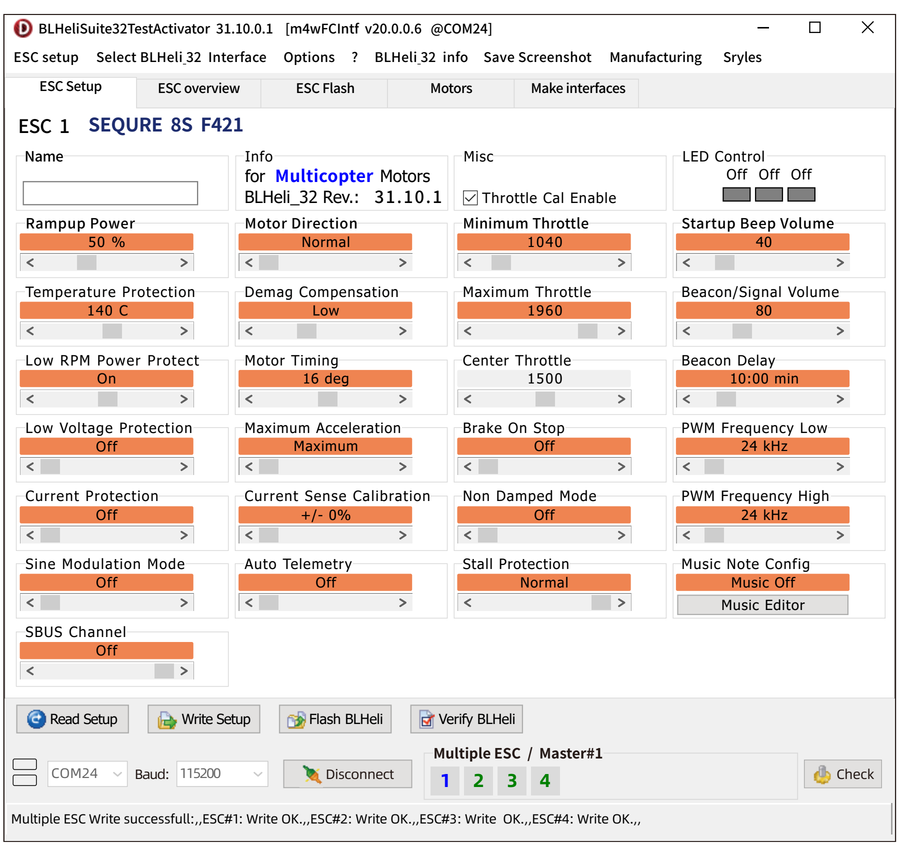

36mm
42.5mm
7mm

重量：17g

4 接线图


M4
M3
M2
M1
TX
CR
GND
VBAT
MOTOR 4
MOTOR 2
MOTOR 3
MOTOR 1
M4
M2
M3
M1
–
+
35V 680uF
SEQURE
(Battery 2-8S)

5 电调参数


BLHeliSuite32TestActivator 31.10.0.1 [m4wFCIntf v20.0.0.6 @COM24]

ESC setup
Select BLHeli_32 Interface
Options
?
BLHeli_32 info
Save Screenshot
Manufacturing
Sryles
ESC Setup
ESC overview
ESC Flash
Motors
Make interfaces

ESC 1 SEQURE 8S F421

Name

Info
for Multicopter Motors
BLHeli_32 Rev.: 31.10.1

Misc
☑ Throttle Cal Enable

LED Control
Off Off Off

Rampup Power: 50 %
Motor Direction: Normal
Minimum Throttle: 1040
Startup Beep Volume: 40
Temperature Protection: 140 C
Demag Compensation: Low
Maximum Throttle: 1960
Beacon/Signal Volume: 80
Low RPM Power Protect: On
Motor Timing: 16 deg
Center Throttle: 1500
Beacon Delay: 10:00 min
Low Voltage Protection: Off
Maximum Acceleration: Maximum
Brake On Stop: Off
PWM Frequency Low: 24 kHz
Current Protection: Off
Current Sense Calibration: +/- 0%
Non Damped Mode: Off
PWM Frequency High: 24 kHz
Sine Modulation Mode: Off
Auto Telemetry: Off
Stall Protection: Normal
Music Note Config: Music Off
SBUS Channel: Off

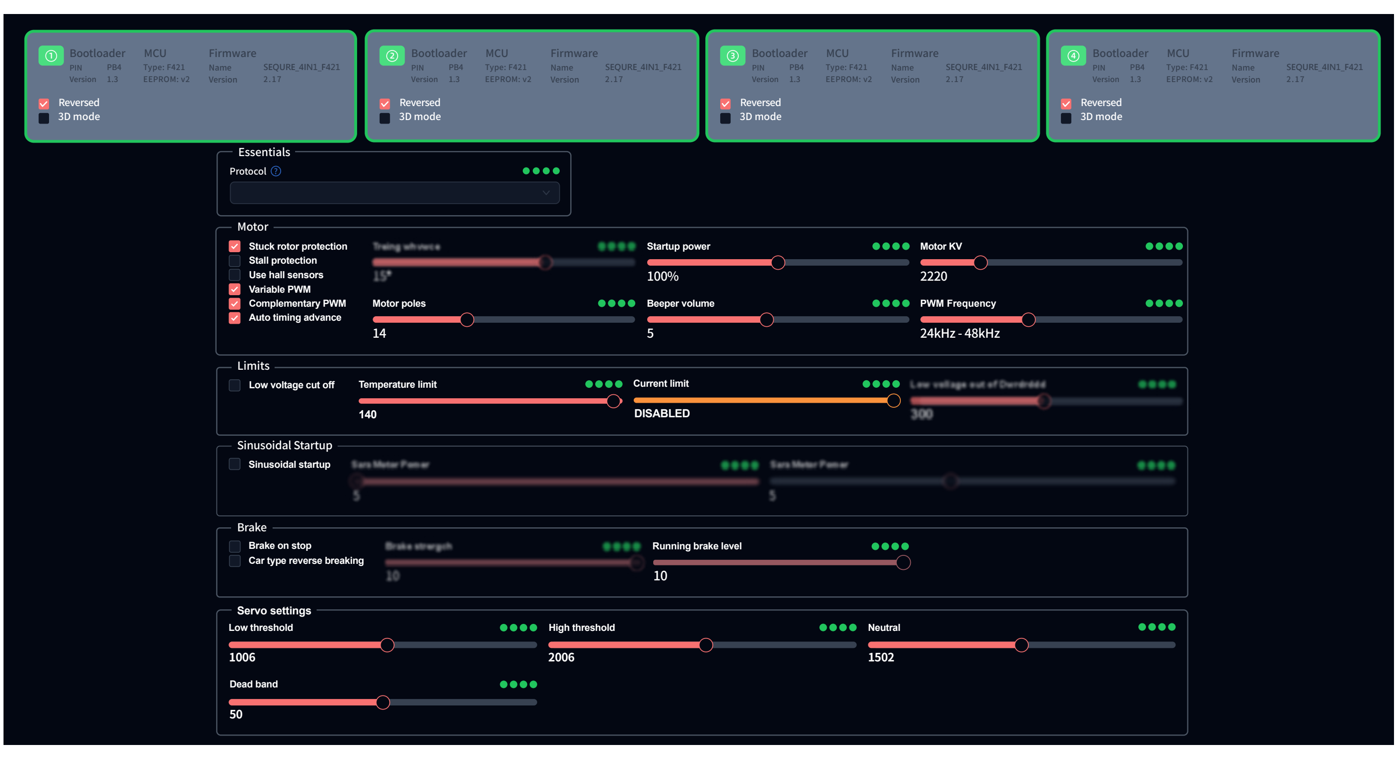
Music Editor
Read Setup
Write Setup
Flash BLHeli
Verify BLHeli
Disconnect
Check

COM24 Baud: 115200

Multiple ESC / Master#1
1 2 3 4

Multiple ESC Write successfull:,,ESC#1: Write OK.,,ESC#2: Write OK.,,ESC#3: Write OK.,,ESC#4: Write OK.,,

BLHeli_32固件可视化编程调参界面


① ② ③ ④ Bootloader PIN PB4 Version 1.3 | MCU Type: F421 EEPROM: v2 | Firmware Name SEQURE_4IN1_F421 Version 2.17
☑ Reversed
■ 3D mode

Essentials
Protocol

Motor
☑ Stuck rotor protection
☐ Stall protection
☐ Use hall sensors
☑ Variable PWM
☑ Complementary PWM
☑ Auto timing advance
Startup power: 100%
Motor KV: 2220
Motor poles: 14
Beeper volume: 5
PWM Frequency: 24kHz - 48kHz

Limits
☐ Low voltage cut off
Temperature limit: 140
Current limit: DISABLED

Sinusoidal Startup
☐ Sinusoidal startup

Brake
☐ Brake on stop
☐ Car type reverse breaking
Running brake level: 10

Servo settings
Low threshold: 1006
High threshold: 2006
Neutral: 1502
Dead band: 50

AM32固件可视化编程调参界面
调参网址：am32.ca

6 保修条款

(1)自购机之日起3个月内出现非人为因素造成的性能故障，厂家提供免费修理服务。3个月后，修理不收取人工费，若更换配件，费用需用户承担。
(2)一切人为损坏，自行拆机，使用不当造成的故障，不在保修范围。

7 安全须知 ⚠

- ●请勿刷写非官方指定的固件，以免损坏电调。
- ●请不要超过ESC工作电压，电流范围使用。
- ●请供电前检查并确保正负极性无误，才能通电使用。
- ●请在符合当地法规和确保安全情况下使用产品。
- ●请使用符合安全标准的配件，存放在儿童不容易接触的地方。

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